

ELEC5305 — Project Progress Submission

Project: When Can Pouring-Sound Fill-Level Estimates Be Trusted?

Student: Yueyao Lyu

Date: 8 October 2026

GitHub Pages: <https://lyu-ui.github.io/elec5305-project-540568813/>

Repository: <https://github.com/lyu-ui/elec5305-project-540568813>

Revised research focus

My project investigates whether interpretable acoustic features can help identify when physics-based pouring-sound liquid-level estimates are reliable or unreliable. In response to feedback, I shifted the focus from a four-class fill-level classifier to an analysis of conditions in which the existing approach succeeds or fails.

What I have achieved

I collected two public recordings and developed a MATLAB FFT/STFT baseline. The prior experimental report includes controlled tests of analysis windows, overlap, resampling, filtering, added noise and signal reconstruction. Five acoustic descriptors—band spectral maximum, centroid, spread, roll-off and RMS—were extracted. A new script calculates candidate resonance-reliability descriptors: ridge prominence, competing-peak ratio, peak jumps and spectral entropy.

Preliminary result and limitation

In the original 40 ms / 50% overlap STFT analysis, the median band maximum changed from 503.9 Hz (early) to 1593.8 Hz (late) in recording A and from 468.8 Hz to 996.1 Hz in recording B. These are not validated fundamental-pitch measurements, and the time thirds are not measured liquid levels. Both clips were recorded with the same container, and a continuous physical-estimation error has not yet been calculated.

Planned work

Reproduce the released Sound of Water pretrained model; analyse reference physical prediction errors on official seen/unseen container splits; test the relationship between interpretable STFT features and error; assess a simple reliability score using selective prediction; and discuss typical acoustic failure modes.

GitHub evidence

The project website and repository include the research question, existing MATLAB code, previous experimental results, dataset provenance, the present limitations and the next development milestones. I welcome feedback on the revised research direction and evaluation plan.